4 Chapter Four: REGRESSION AND CORRELATIONS

4.1.1 LINEAR REGRESSION

When two varieties x, y, whose measurements have been shown on a graph and look like the following figures below:

It is natural to try to describe the linear, or approximately linear, relation between them by an equation, the **linear regression** equation. This expresses one variate, the **dependent** variable *y*, in terms of the other, the **independent** variable *x*.

We can properly write $y = b_0 + b_1 x$ to show how *y* alters when *x* is altered, where "b_0" is the **intercept**, or that value taken by *y* when *x* = 0, while "b_1" is the **slope** or gradient of the line, the average increase in *y* for unit increase in *x*. We can choose, and fix, the values of *x* to use in an experiment designed to study this relation, but we have no control over the resulting *y*.

4.1.2 ORDINARY LEAST SQUARES METHOD (OLS)

Is a technique of fitting the "best" curve to a set of observations $(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$. If the points are plotted on a rectangular co-ordinate system, it is possible to visualize the relationship between Y and X. The plotted set of points is called a **scatter diagram**.

The process of estimating the parameters of the equation y = f (x) using the method of least squares is referred to as **regression**.

4.1.3 Simple Linear Regression Analysis

Given the data $(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$, the problem of linear regression analysis is to find the estimates of "a" and "b" in the equation $y = a + bx$

which best fit the data. The line $\hat{y} = \hat{a} + \hat{b}x$ (where $\hat{a}$ and $\hat{b}$ are the least squares estimates of a and b respectively) is called the least squares regression line of Y on X.

4.1.3.1 Note

We assume that the values of the independent variable (i.e. X) are obtained without error. The predicted value of *y* at a given value of *x* is given by $\hat{y} = \hat{a} + \hat{b}x$.

In general, $\hat{y}_i \neq y_i$.

Define $d_i = y_i - y_i^{(p)}$ (is the difference between the observed value and the predicted value)

The approximating line L is the best fitting line (or the least squares line if $\sum_{i=1}^{n} d_i^2$ is a minimum)

4.2 Method: To find the least squares regression line of y on x

Given the data $(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$, we minimize:

$$S = \sum_{i=1}^{n} d_i^2$$

$$= \sum_{i=1}^{n} \left(y_i - y_i^{(p)}\right)^2 = \sum_{i=1}^{n} \left[y_i - (b_0 + b_1 x_i)\right]^2 = \sum_{i=1}^{n} \left(y_i - b_0 - b_1 x_i\right)^2$$

The least squares estimates of b_0 and b_1 satisfy:

$$\frac{\partial S}{\partial b_0} = 0 \quad and \quad \frac{\partial S}{\partial b_1} = 0 \;\} \; for\ a\ \min imum$$

$$\frac{\partial S}{\partial b_0} = -2\sum \left(y_i - b_0 - b_1 x_i\right) = 0 \quad \cdots\cdots\cdots\cdots (i)$$

$$\frac{\partial S}{\partial b_1} = -2\sum \left(y_i - b_0 - b_1 x_i\right) x_i = 0 \quad \cdots\cdots\cdots\cdots (ii)$$

Solving equations (i) and (ii) simultaneously, we get

$$\hat{b}_1 = \frac{n\sum x_i y_i - \left(\sum x_i\right)\left(\sum y_i\right)}{n\sum x_i^2 - \left(\sum x_i\right)^2} \quad \text{and} \quad \hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$$

The line $\hat{y} = \hat{b}_0 + \hat{b}_1 x$ is called the least squares regression line of Y on X since it is estimated from X.

4.3 Example

For the given data below, calculate the least squares regression line of Y on X. Estimate the value of Y when X = 12, X = 15.

Mean Temp. (X)°C	7	5	5	9	10	15	18	19
Energy Consumption (Y)	29	30	34	27	23	25	19	18

4.4 Solution

x	**y**	**xy**	**x^2**
7	29	203	49
5	30	150	25
5	34	170	25
9	27	243	81
10	23	230	100
15	25	375	225
18	19	342	324
19	18	342	361
$\sum x = 88$	$\sum y = 205$	$\sum xy = 2055$	$\sum x^2 = 1190$

$$\hat{b}_1 = \frac{n\sum x_i y_i - \left(\sum x_i\right)\left(\sum y_i\right)}{n\sum x_i^2 - \left(\sum x_i\right)^2}$$

$$= \frac{(8)(2055) - (88)(205)}{(8)(1190) - 88^2} = -0.9$$

$$\hat{b}_0 = \bar{y} - \hat{b}_1\bar{x} = \frac{205}{8} - (-0.9)(88/8) = 35.525$$

$$\hat{y} = \hat{b}_0 + \hat{b}_1 x = 35.525 - 0.9x$$

The predicted energy consumption when the temperature is 12^0C is given by:

$$\hat{y} = 35.525 - 0.9x = 35.525 - 0.9(12) = 24.73 \; units$$

When x = 15^0C; $\hat{y} = 35.525 - 0.9x = 35.525 - 0.9(15) = 22.03 \; units$

Note the actual value observed is 25 units, $\therefore d = 25 - 22.03 = 2.97$

4.4.1 Simple Linear Regression Model

We have a set of data $(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)$ where it is assumed that for fixed X; the conditional density of Y given X is normally distributed with $Mean = b_0 + b_1 x$ and $Variance = \sigma^2$, where b_0, b_1, and σ^2 are unknown constants.

i.e. $Y/X \sim N(b_0 + b_1 x, \sigma^2)$

$E(Y/X) = b_0 + b_1 x$

The linear relationship between Y and X can be made stochastic by adding an error term μ. That is; $y = b_0 + b_1 x + \mu$. This is called a Simple Linear Regression Model. Normal linear regression analysis is concerned with the:

(i) Estimation of the parameters b_0, b_1, and σ^2 (if μ is unknown).

(ii) Testing of hypothesis concerning these parameters.

(iii) Prediction based on the estimated regression model.

4.5 Classical assumptions of linear regression model

(i) $\mu \sim N(0, \sigma^2)$

(ii) X can be measured without error, i.e. $E(Y/X) = b_0 + b_1 x$ or

$Y/X \sim N(b_0 + b_1 x, \sigma^2)$

Note that these assumptions are necessary for testing hypothesis but <u>not</u> for obtaining the estimates of b_0 and b_1.

We have seen earlier that the least squares estimates of b_0 and b_1 are given by:

$$\hat{b}_1 = \frac{n\sum x_i y_i - \left(\sum x_i\right)\left(\sum y_i\right)}{n\sum x_i^2 - \left(\sum x_i\right)^2} \quad \text{and} \quad \hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x}$$

The estimated regression equation is given by $\hat{y} = \hat{b}_0 + \hat{b}_1 x$. $\hat{y}$ is called the predicted value of Y corresponding to a given value of X.

$\hat{b}_1$ can also be expressed in each of the following alternative forms:

$$(i)\ \hat{b}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \quad (ii)\ \hat{b}_1 = \frac{\sum (x_i - \bar{x}) y_i}{\sum (x_i - \bar{x})^2} \quad (iii)\ \hat{b}_1 = \frac{\sum (y_i - \bar{y}) x_i}{\sum (x_i - \bar{x})^2}$$

4.5.1 Test of Significance of the parameter estimates

To test for the significance of b_0 and b_1, we require the mean and variance of $\hat{b}_0$ *and* $\hat{b}_1$.

Model: $y_i = b_0 + b_1 x_i + \mu_i$ *where* $\mu_i \sim N(0, \sigma^2)$ $i = 1, 2, \cdots, n$

4.5.1.1.1 Assumptions

$(i)\ \mu_i \sim N(0, \sigma^2),\ i = 1, 2, \cdots, n$

$(ii)\ Cov(\mu_i, \mu_j) = 0$ *for* $i \neq j$, *i.e. the errorterms are uncorrelated*

$(iii)\ x_i$*'s* are fixed known values.

4.5.1.2 Mean and Variance of $\hat{b}_0$ *and* $\hat{b}_1$

$$E(\hat{b}_1)=E\left[\frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}\right]=b_1\frac{\sum(x_i-\bar{x})^2}{\sum(x_i-\bar{x})^2}=b_1$$

Therefore $\hat{b}_1$ is an unbiased estimate of b_1.

$$E(\hat{b}_0)=E(\bar{y}-\hat{b}_1\bar{x})=b_0+b_1\bar{x}+E(\bar{\mu})-\bar{x}E(\hat{b}_1)$$
$$=b_0+b_1\bar{x}+0-b_1\bar{x}=b_0$$

$$\therefore E(\hat{b}_0)=b_0$$

Therefore $\hat{b}_0$ is an unbiased estimate of b_0

4.5.1.3 To find variance of $\hat{b}_1$ (assuming σ^2 is known)

$$Var(\hat{b}_1)=Var\left[\frac{\sum(x_i-\bar{x})y_i}{\sum(x_i-\bar{x})^2}\right]=\frac{\sigma^2}{\sum(x_i-\bar{x})^2}$$

Note: $\hat{b}_0$ *and* $\hat{b}_1$ are normally distributed random variables because they are linear functions of normal variables $y_i^{'s}$. Therefore:

$$\hat{b}_1 \text{ is } N\left(b_1,\ \frac{\sigma^2}{\sum(x_i-\bar{x})^2}\right)$$

4.5.1.4 To find Variance of $\hat{b}_0$ (assuming σ^2 is known)

$$Var(\hat{b}_0)=Var(\bar{y}-\hat{b}_1\bar{x})$$
$$=Var\left(b_0+b_1\bar{x}+\bar{\mu}-\hat{b}_1\bar{x}\right)$$

which upon reduction becomes

$$Var\left(\hat{b}_0\right)=\sigma^2\left[\frac{1}{n}+\frac{\bar{x}^2}{\sum(x_i-\bar{x})^2}\right]$$

i.e. $\hat{b}_0 \sim N\left(b_0, \sigma^2\left[\frac{1}{n} + \frac{\bar{x}^2}{\sum(x_i - \bar{x})^2}\right]\right)$

Since σ^2 is usually unknown, $S^2 = \frac{\sum(y_i - \hat{y}_i)^2}{n-2}$ is used as an unbiased estimate of σ^2.

4.6 Theorem

The random variable $V = \frac{\sum(y_i - \hat{y}_i)^2}{\sigma^2}$ has a χ^2 - distribution with (n - 2) d.f.

4.6.1 To construct a T – test

Recall that if U *is* $N(0,1)$ and V *is* $\chi^2(r)$ and further that U and V are independent, then the random variable $T = \frac{U}{\sqrt{V/r}}$ has a students t – distribution with r – degrees of freedom.

In our problem;

$U = \frac{\hat{b}_1 - b_1}{\sqrt{\sigma^2 / \sum(x_i - \bar{x})^2}}$ *is* $N(0,1)$ and

$V = \frac{\sum(y_i - \hat{y}_i)^2}{\sigma^2} = \frac{(n-2)S^2}{\sigma^2}$ is distributed as $\chi^2(n-2)$

Assuming that U and V are independent,

$$T = \frac{U}{\sqrt{V/r}} = \frac{\dfrac{\hat{b}_1 - b_1}{\sqrt{\dfrac{\sigma^2}{\sum (x_i - \bar{x})^2}}}}{\sqrt{\dfrac{(n-2)S^2}{\sigma^2(n-2)}}}$$

$$= \frac{(\hat{b}_1 - b_1)\sqrt{\sum (x_i - \bar{x})^2}}{S}$$ has a students t – distribution with (n - 2) d.f.

Note that $Var(\hat{b}_1) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$. If σ^2 is unknown we replace it by

$S^2 = \frac{\sum (y_i - \hat{y}_i)^2}{n-2}$.

The estimated variance of $\hat{b}_1 = \frac{S^2}{\sum (x_i - \bar{x})^2} = S^2_{\hat{b}_1}$. $S^2_{\hat{b}_1}$ is an unbiased estimate

of $Var(\hat{b}_1)$. $S^2_{b_1} = \sqrt{\frac{S^2}{\sum (x_i - \bar{x})^2}} = \frac{S}{\sqrt{\sum (x_i - \bar{x})^2}}$ is called the standard error of

$\hat{b}_1$.

Therefore $T = \frac{\hat{b}_1 - b_1}{S_{\hat{b}_1}}$.

To test the hypothesis $\begin{matrix} H_0 : b_1 = B_1 \\ H_1 : b_1 \neq B_1 \end{matrix}$ V/S the test is to compute $T = \frac{\hat{b}_1 - B_1}{S_{\hat{b}_1}}$

which is distributed as $t_{(n-2)}$ when H_0 is true. Compare this value of T with the tabulated values of "t" with (n - 2) d.f. at α – level of significance.

If $|T| > t_{\alpha/2}$ (*tabulated*) reject H_0.

In particular if $B_1 = 0$ we say that b_1 is significant if H_0 is rejected.

To test for the hypothesis $\begin{matrix} H_0 : b_1 = B_1 \\ H_1 : b_1 > B_1 \end{matrix}$ V/S, reject H_0 if $T > t_\alpha$.

In a similar manner to test for the hypothesis $\begin{matrix} H_0 : b_0 = B_0 \\ H_1 : b_0 \neq B_0 \end{matrix}$ V/S, the test is

to compute $T = \dfrac{\hat{b}_0 - B_0}{S_{\hat{b}_0}}$ which is distributed as $t_{(n-2)}$ when H_0 is true,

where

$S_{\hat{b}_0} = S\sqrt{\left[\dfrac{1}{n} + \dfrac{\bar{x}^2}{\sum(x_i - \bar{x})^2}\right]}$ is the standard error of $\hat{b}_0$ and $S = \sqrt{\dfrac{\sum(y_i - \hat{y}_i)^2}{n-2}}$.

4.6.2 Goodness of fit and correlation coefficient

The closer the observations fall to the regression line, the greater is the variation in *y* explained by the regression equation.

The degree of this variation is measured by the correlation coefficient.

We have $\hat{y} = \hat{a} + \hat{b}x$ where $\hat{y}$ is the predicted value of y.

4.7 Definitions

(i) The quantity $\sum_{i=1}^{n}(y_i - \bar{y})^2$ is called the total sum of squares (T.S.S) or total variation.

(ii) The quantity $\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$ is called the regression sum of squares (R.S.S) or the explained variation.

(iii) The quantity $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$ is called the error sum of squares (E.S.S) or the unexplained variation.

4.8 Theorem

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$$

Hence T.S.S. = E.S.S. + R.S.S

4.8.1 Coefficient of determination (R^2)

Is defined as the proportion of the total variation (T.S.S) in Y explained by the regression Y on X (R.S.S)

4.9 Definition

$$R^2 = \frac{R.S.S.}{T.S.S.} = 1 - \frac{E.S.S.}{T.S.S.}$$

$$\therefore\ R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2}$$

Note:

(i) $0 < R^2 < 1$

(ii) If $R^2 = 1$ all the sample points lie on the regression line (there is a perfect linear relationship between Y and X).

(iii) If $R^2 = 0$ all the sample points lie on a horizontal line or a circle.

4.9.1 Linear Correlation Coefficient (r)

4.10 Definition

$$r = \sqrt{R^2}$$

$$= \sqrt{1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2}}$$

Which on reducing becomes;

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} = \frac{Cov(x, y)}{\sqrt{Var(x) Var(y)}}$$

This formula automatically gives the correct sign associated with r.

Note that $-1 \le r \le 1$.

The correlation coefficient measures the goodness of fit on the regression equation actually assumed. In our case r measures the degree of linear relationship between Y and X.

4.10.1 Interpretation of r

(i) If r = 0, there exists no linear relationship between Y and X.

(ii) If r = 1, there exists a perfect direct linear relationship between Y and X. i.e. the set of data points $\{(x_i, y_i);\ i=1,2,\cdots,n\}$

(iii) If r = -1, there exists a perfect inverse linear relationship between Y and X.

(iv) If r > 0, there is some direct linear relationship between Y and X.

(v) If r < 0, there is some inverse linear relationship between Y and X.

4.10.2 Confidence interval for b_0 and b_1

We can also obtain the (1 - □)100% confidence interval for b_0 and b_1 given by:

$$\Pr\left[-t_{\alpha/2} < t < t_{\alpha/2}\right] = 1-\alpha \quad (\alpha \text{ specified})$$

$$\Rightarrow \Pr\left[-t_{\alpha/2} < \frac{\hat{b}_0 - B_0}{S_{\hat{b}_0}} < t_{\alpha/2}\right] = 1-\alpha$$

$$\Rightarrow \Pr\left[-t_{\alpha/2} S_{\hat{b}_0} < \hat{b}_0 - B_0 < t_{\alpha/2} S_{\hat{b}_0}\right] = 1-\alpha$$

$$\Rightarrow \Pr\left[-\hat{b}_0 - t_{\alpha/2} S_{\hat{b}_0} < -B_0 < -\hat{b}_0 + t_{\alpha/2} S_{\hat{b}_0}\right] = 1-\alpha$$

$$\Rightarrow \Pr\left[\hat{b}_0 - t_{\alpha/2} S_{\hat{b}_0} < B_0 < \hat{b}_0 + t_{\alpha/2} S_{\hat{b}_0}\right] = 1-\alpha$$

4.11 $\therefore$ The (1 - □)100% confidence interval for b_0 is given by

$$\left(\hat{b}_0 - t_{\alpha/2} S_{\hat{b}_0},\ \hat{b}_0 + t_{\alpha/2} S_{\hat{b}_0}\right)$$

4.12 Alternatively, to test;

$$H_0 : \hat{b}_0 = B_0 \quad V/S$$

$$H_0 : \hat{b}_0 \neq B_0$$

we accept the null hypothesis if b_0 is obtained in the interval $\left(\hat{b}_0 - t_{\alpha/2} S_{\hat{b}_0}, \hat{b}_0 + t_{\alpha/2} S_{\hat{b}_0}\right)$. Reject H_0, if b_o is <u>not</u> obtained in the interval.

Similarly, the (1 - □)100% confidence interval for b_1 is given by $\left(\hat{b}_1 - t_{\alpha/2} S_{\hat{b}_1}, \hat{b}_1 + t_{\alpha/2} S_{\hat{b}_1}\right)$

4.13 RESIDUALS

The ith residual is the difference between the observed Y_i and the corresponding fitted value $\hat{Y}_i$. This residual is denoted by e_i and is defined in general as follows:

$e_i = Y_i - \hat{Y}_i$. For the simple regression model, the residual e_i becomes:

$e_i = Y_i - \hat{Y}_i = Y_i - (b_0 + b_1 X_i) = Y_i - b_0 - b_1 X_i$.

The adjusted residuals are given by $\frac{e_i}{s.e.(e_i)} = \frac{Y_i - \hat{Y}_i}{s.e.(Y_i - \hat{Y}_i)}$.

4.14 Example 1

The table below gives the values of Y for given values of X.

X	12	14	10	13	17	12	11	15
Y	5	11	7	8	11	7	6	9

(a) Plot the data. What can you say about the plot of the data?

(b) Estimate the regression equation of Y on X.

(c) If the assumptions of the linear model are valid, test for the statistical significance of the parameter estimates at 5% level.

(d) What are the predicted values?

(e) Compute the residuals and the adjusted residuals and interpret.

(f) Plot a graph of residuals against the predicted values.

(g) Compute the 95% confidence intervals for b_0 and b_1.

4.14.1.1 Solution

Model: $y_i = b_0 + b_1 x_i + \mu_i$ *where* $\mu_i \sim N(0, \sigma^2)$ $i = 1, 2, \cdots, n$, $Cov(\mu_i, \mu_j) = 0$ *for* $i \neq j$

$$\bar{x} = \frac{\sum x}{8} = 13, \quad \bar{y} = \frac{\sum y}{8} = 8$$

Table:

x	y	$(x-\bar{x})$	$(x-\bar{x})^2$	$(y-\bar{y})$	$(y-\bar{y})^2$	$(x-\bar{x})(y-\bar{y})$
12	5	-1	1	-3	9	3
14	11	1	1	3	9	3
10	7	-3	9	-1	1	3
13	8	0	0	0	0	0
17	11	4	16	3	9	12
12	7	-1	1	-1	1	1
11	6	-2	4	-2	4	4
15	9	2	4	1	1	2
			$\sum(x-\bar{x})^2 = 36$		$\sum(y-\bar{y})^2 = 34$	$\sum(x-\bar{x})(y-\bar{y}) =$ 28

$$\hat{b}_1 = \frac{n\sum x_i y_i - \left(\sum x_i\right)\left(\sum y_i\right)}{n\sum x_i^2 - \left(\sum x_i\right)^2} = \frac{28}{36} = 0.7778 \quad \hat{b}_0 = \bar{y} - \hat{b}_1 \bar{x} = 8 - (0.7778)13 = -2.111$$

The estimated regression equation is given by $\hat{y} = \hat{b}_0 + \hat{b}_1 x = -2.111 + 0.7778x$.

To test for the significance of b_1

We want to test the hypothesis $\begin{matrix} H_0 : b_1 = 0 \\ H_1 : b_1 \neq 0 \end{matrix}$ *Versus*. The test is to compute

$t = \dfrac{\hat{b}_1}{S_{\hat{b}_1}}$ which is distributed as $t_{(n-2)}$ when H_0 is true, where

$$S_{\hat{b}_1} = \frac{S}{\sqrt{\sum (x_i - \bar{x})^2}}.$$

$$S_{\hat{b}_1} = \frac{S}{\sqrt{\sum (x_i - \bar{x})^2}} = S\sqrt{\frac{1}{36}} = \frac{S}{6}.$$

We want to evaluate S^2.

$S^2 = \dfrac{\sum (y_i - \hat{y}_i)^2}{n-2} = \dfrac{\sum (y_i - \hat{y}_i)^2}{6}$. To evaluate $\sum (y_i - \hat{y}_i)^2$, we use the relationship $r = 1 - \dfrac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2} = 1 - \dfrac{\sum (y_i - \hat{y}_i)^2}{34}$ ……(*)

also $r^2 = \dfrac{\left[\sum (x_i - \bar{x})(y_i - \bar{y})\right]^2}{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2} = \dfrac{28^2}{36 \times 34} = 0.6405$

Therefore from equation (*) above; $0.6405 = 1 - \dfrac{\sum (y_i - \hat{y}_i)^2}{34}$

$$\Rightarrow \frac{\sum (y_i - \hat{y}_i)^2}{34} = 1 - 0.6405 \Rightarrow \sum (y_i - \hat{y}_i)^2 = 12.22$$

Hence $S^2 = \dfrac{\sum (y_i - \hat{y}_i)^2}{6} = \dfrac{12.22}{6} = 2.037 \Rightarrow S = \sqrt{2.037} = 1.427$

Therefore $S_{\hat{b}_1} = \dfrac{S}{6} = \dfrac{1.427}{6} = 0.238$

$$\therefore t = \frac{\hat{b}_1}{S_{\hat{b}_1}} = \frac{0.7778}{0.238} = 3.27 \Rightarrow |t| = 3.27$$

But $t_{0.025}(6) = 2.45$

4.14.1.2

4.14.1.3 Conclusion

$|t| = 3.27 > t_{0.025}(6) = 2.45$, we therefore reject the null hypothesis H_0 at 5% level of significance. That is, from the sample observations, b_1 is significantly different from zero at 5% level of significance.

To test for the significance of b_0

We want to test the hypothesis $\begin{matrix} H_0 : b_0 = 0 \\ H_1 : b_0 \neq 0 \end{matrix}$ *Versus*. The test is to compute

$t = \dfrac{\hat{b}_0}{S_{\hat{b}_0}}$ which is distributed as $t_{(n-2)}$ when H_0 is true, where

$$S_{\hat{b}_0} = S\sqrt{\left[\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2}\right]} = 1.427\sqrt{\left[\frac{1}{8} + \frac{13^2}{36}\right]} = 3.1327.$$

$$\therefore\ t = \frac{\hat{b}_0}{S_{\hat{b}_0}} = \frac{-2.111}{3.1327} = -0.673 \ \Rightarrow\ |t| = 0.673$$

4.14.1.4 Conclusion

$|t| = 0.673 < t_{0.025}(6) = 2.45$, we therefore accept the null hypothesis H_0 at 5% level of significance. That is, from the sample observations, b_0 is not significantly different from zero at 5% level of significance.

8 ©The 95% confidence interval for b_0 is given by

$$\left(\hat{b}_0 - t_{\alpha/2}S_{\hat{b}_0},\ \hat{b}_0 + t_{\alpha/2}S_{\hat{b}_0}\right) = -2.111 \pm (2.45) \times (3.1327) = [-9.786,\ 5.564]$$

The 95% confidence interval for b_1 is given by

$$\left(\hat{b}_1 - t_{\alpha/2}S_{\hat{b}_1},\ \hat{b}_1 + t_{\alpha/2}S_{\hat{b}_1}\right) = 0.7778 \pm (2.45) \times (0.238) = [0.1947,\ 1.361]$$

Exercise

(1) A random sample of 8 men was taken and the following information was gathered:

Height	62	64	58	63	65	59	60	61
Age	20	25	15	21	24	16	17	18

(a) Regress height on age and find the regression equation.

(b) If the assumptions of the linear model are valid, test for the statistical significance of the parameter estimates of slope at 5% level.

$[\hat{y} = -69.821 + 1.4524x,\ S = 0.949,\ b_1 : t = 9.948,\ \text{reject } H_0,\ b_0 : t = -7.748,\ \text{reject } H_0\ ]$

(2) The following table shows the assessed values (in thousands dollars) and the selling prices (in thousands dollars) of eight houses, constituting a random sample of all houses sold recently in a metropolitan area:

Assessed value	70.3	102	62.5	74.8	57.9	81.6	110.4	88
Selling price	114.4	169.3	106.2	125	99.8	132.1	174.2	143.5

(a) Fit a least squares line that will enable us to predict the selling price of a house in that metropolitan area in terms of its assessed value.

(b) Test the hypothesis $H_0 : b_1 = 1.3$ *Versus* $H_1 : b_1 > 1.3$ at 5% level of significance.

$[\hat{y} = 12.247 + 1.493x, t = 3.41,\ \text{reject } H_0\ ]$

(3) The following data show the advertising expenses (in thousands dollars) and the net operating profits (in thousands dollars) in a random sample of six drugstores:

Advertising expenses	1.5	1.0	2.8	0.4	1.3	2.0
Net operating profit	3.6	2.8	5.4	1.9	2.9	4.3

(a) Fit a least squares line that will enable us to predict the net operating profit in terms of advertising expenses.

(b) Test the hypothesis $H_0 : b_0 = 0.8$ *Versus* $H_1 : b_0 > 0.8$ at the 1% level of significance.

$[\hat{y} = 1.259 + 1.483x, t = 3.10,\ accept\ H_0]$